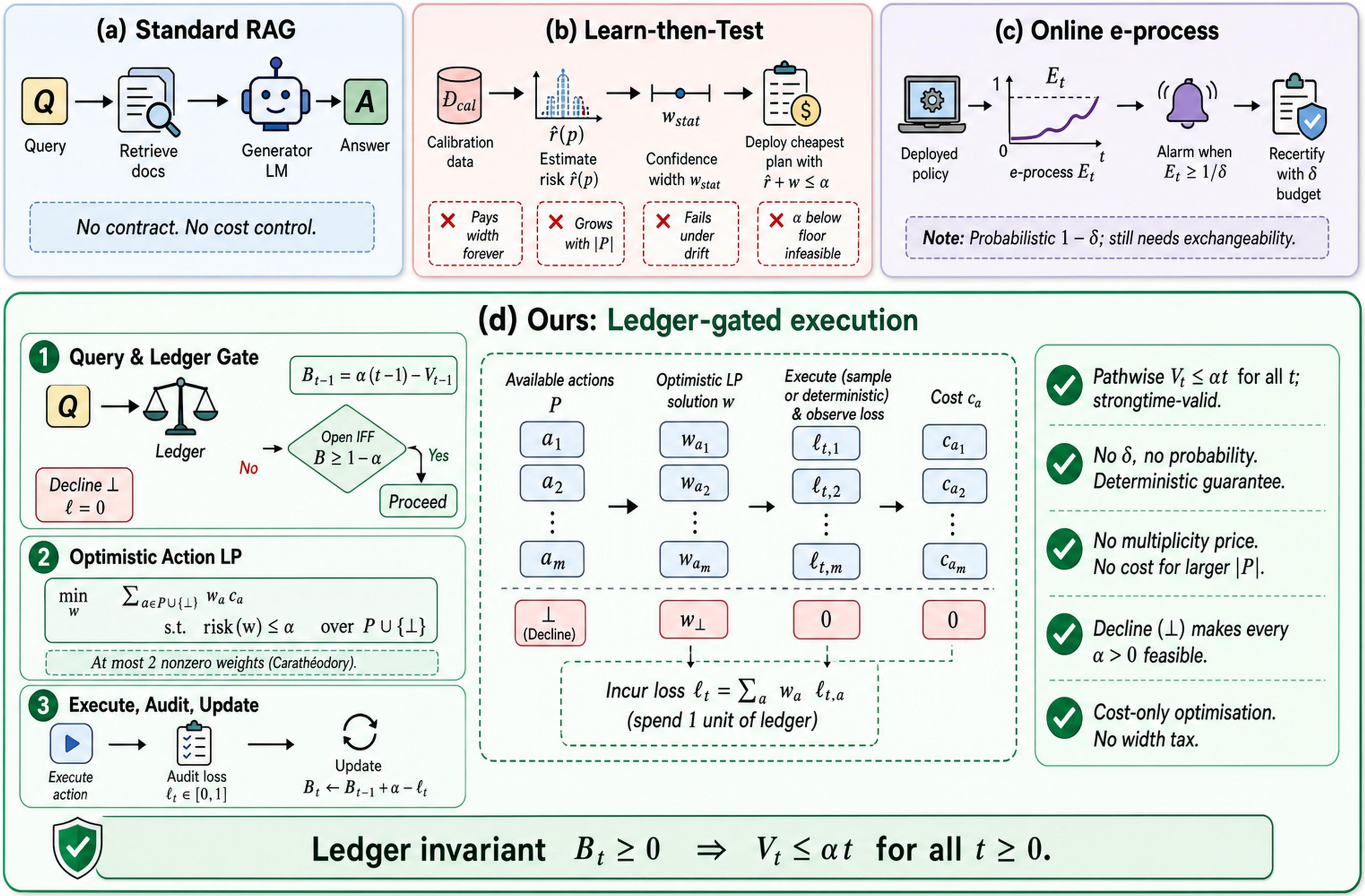
(a) Standard RAG
Q
Query
Retrieve docs
Generator LM
A
Answer
No contract. No cost control.
(b) Learn-then-Test
$\mathcal{D}_{cal}$
Calibration data
$\hat{r}(p)$
Estimate risk $\hat{r}(p)$
w_{stat}
Confidence width w_{stat}
Deploy cheapest plan with $\hat{r} + w \leq \alpha$
Pays width forever
Grows with $|P|$
Fails under drift
α below floor infeasible
(c) Online e-process
Deployed policy
E_t
1
0
t
e-process E_t
Alarm when $E_t \geq 1/\delta$
Recertify with δ budget
Note: Probabilistic $1-\delta$; still needs exchangeability.
(d) Ours: Ledger-gated execution
1 Query & Ledger Gate
Q
Ledger
$B_{t-1} = \alpha(t-1) - V_{t-1}$
Open IFF $B \geq 1-\alpha$
No
Yes
Decline $\perp$ $\ell = 0$
Proceed
2 Optimistic Action LP
$\min_w \sum_{a \in P \cup \{\perp\}} w_a c_a$ s.t. $\mathrm{risk}(w) \leq \alpha$ over $P \cup \{\perp\}$
At most 2 nonzero weights (Carathéodory).
3 Execute, Audit, Update
Execute action
Audit loss $\ell_t \in [0,1]$
Update $B_t \leftarrow B_{t-1} + \alpha - \ell_t$
Available actions P
Optimistic LP solution w
Execute (sample or deterministic) & observe loss
Cost c_a
a_1 w_{a_1} $\ell_{t,1}$ c_{a_1}
a_2 w_{a_2} $\ell_{t,2}$ c_{a_2}
a_m w_{a_m} $\ell_{t,m}$ c_{a_m}
$\perp$ (Decline) $w_\perp$ 0 0
Incur loss $\ell_t = \sum_a w_a \ell_{t,a}$ (spend 1 unit of ledger)
Pathwise $V_t \leq \alpha t$ for all t; strongtime-valid.
No δ, no probability. Deterministic guarantee.
No multiplicity price. No cost for larger $|P|$.
Decline ($\perp$) makes every $\alpha > 0$ feasible.
Cost-only optimisation. No width tax.
Ledger invariant $B_t \geq 0 \Rightarrow V_t \leq \alpha t$ for all $t \geq 0$.